Problem Set 10

1. Polyphase decomposition

Let $h[n] = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $M = 3$.

(a) Find $e_0[r], e_1[r]$, and $e_2[r]$ using $e_k[r] = h[rM + k]$.

(b) Write $H(z)$ in polyphase form $H(z) = \sum_k z^{-k} E_k(z^M)$.

(c) Explain how the original tap ordering is recovered when the phases are recombined.

2. Where the decimation savings comes from

A 48-tap FIR anti-aliasing filter precedes a downsampler by $M = 6$. The input rate is 96 kHz.

(a) Estimate the multiply rate for the naïve filter-then-downsample implementation.

(b) Estimate the multiply rate for an ideal polyphase implementation.

(c) State the approximate speedup and explain why the number of multiplies per retained output is not reduced by the same factor.

3. First Noble Identity and its limitation

Let $G(z) = 1 + 2z^{-1}$ and $M = 2$. Use $x[n] = \delta[n]$.

(a) Compute the output of $G(z^2)$ followed by $\downarrow 2$.

(b) Compute the output of $\downarrow 2$ followed by $G(z)$. Verify equivalence.

(c) Now replace $G(z^2)$ in the left system by the arbitrary filter $H(z) = 1 + 2z^{-1}$. Show that $H(z) \rightarrow \downarrow 2$ is not equivalent to $\downarrow 2 \rightarrow H(z)$. Explain the lesson learned.

4. Efficient interpolation phases

Let $h[n] = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and interpolation factor $L = 4$.

(a) Find the four polyphase impulse responses.

(b) Which phase produces output samples $y[4m + 2]$?

(c) If the original FIR has $N = 40$ taps, compare the approximate multiply rates of naïve zero-insertion + full-rate filtering and polyphase interpolation.

5. Haar analysis/synthesis bank

Use the pairwise Haar transform

$$a[m] = \frac{1}{\sqrt{2}}(x[2m] + x[2m + 1]) \quad \text{and} \quad d[m] = \frac{1}{\sqrt{2}}(x[2m] - x[2m + 1]).$$

For $x = \{3, 1, 2, -2\}$:

(a) Compute $a[m]$ and $d[m]$.

(b) Reconstruct x exactly using the synthesis equations.

(c) Verify that input energy equals total subband energy.

6. Perfect reconstruction reasoning (ECEN 5632 Problem)

A two-channel critically sampled analysis/synthesis bank uses analysis filters $H_0(z), H_1(z)$ and synthesis filters $F_0(z), F_1(z)$.

(a) In words, what two things must be true for perfect reconstruction?

(b) Interpret the alias-cancellation condition $F_0(z)H_0(-z) + F_1(z)H_1(-z) = 0$.

(c) Why is “the two filters cover low and high frequencies” not by itself sufficient to guarantee perfect reconstruction?